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Case Study II:
Styrian Health Data

Week 5 Update: Spatial analysis and Model Improvements

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Study Overview & Objectives

Study Focus:

- This study applies spatial analysis to patient-level BP data across Austria to uncover geographic patterns.

Research Questions:

1. Does **blood pressure vary** by **urban-rural typology**?
2. Is there **spatial clustering of hypertension** across districts?
3. Does **regional GDP (NUTS3)** correlate with **blood pressure levels**?

Data Sources

- Patient-level **BP records** linked to Austrian administrative geography (**Bezirk/NUTS3**)
- Open-source regional indicators from `https://www.statistik.at/atlas/`
  - Gross Domestic Product per Capita - NUTS3
  - Divisions according to urban and rural areas

Data Processing Notes

- **Urban–Rural match rate:** Municipality-level matching performed using the Eurostat urban–rural typology CSV.
- Some *Gemeinden* could not be matched directly due to post-2015 Austrian municipal mergers (e.g., Bruck an der Mur + Mürzzuschlag → Bruck-Mürzzuschlag; Hartberg + Fürstenfeld → Hartberg-Fürstenfeld).
- **Patient threshold:** Only districts and NUTS3 regions with ≥ 30 patients were retained for spatial and GDP-related analyses.
- Grey districts/regions on maps indicate missing data: either no patients recorded in that area or unresolved municipality matching.

Data Limitations

- **Urban–Rural match rate:** Municipality-level matching performed using the Eurostat urban–rural typology CSV.
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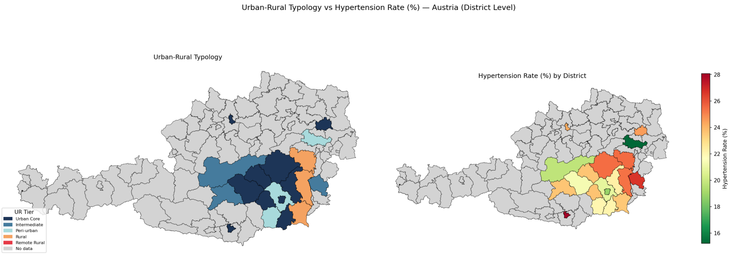
Module 1 — Urban-Rural Typology

- Patients matched to Eurostat urban-rural typology (*Gemeinde* level)
- Five tiers: Urban Core, Intermediate, Peri-urban, Rural, Remote Rural
- District classification based on modal urban-rural tier
- Hypertension:

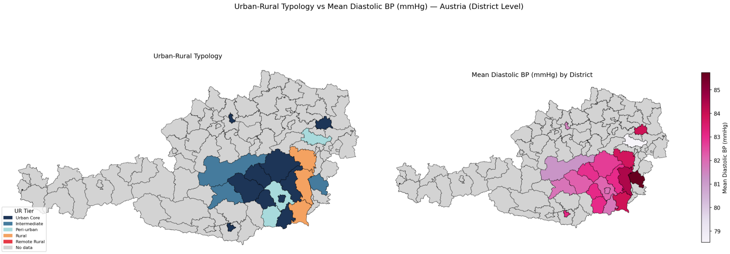
$$\text{SBP} \geq 140 \text{ mmHg}$$

- Spatial weights: Queen contiguity (row-standardised)
- Spatial tests: Moran's I and LISA

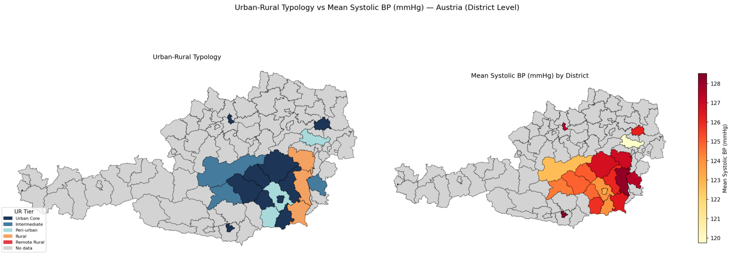
Urban-Rural Typology vs Hypertension



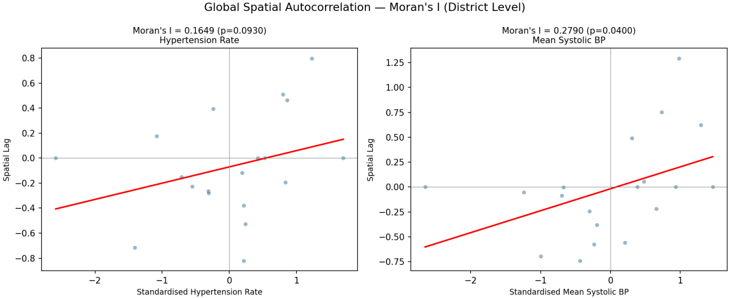
Urban-Rural Typology vs Mean Diastolic BP



Urban-Rural Typology vs Mean Systolic BP

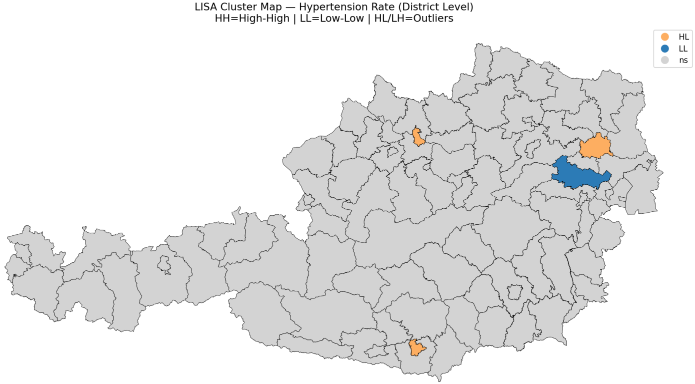


Spatial Autocorrelation Analysis (Moran's I)



- Urban–rural typology alone does not explain the spatial pattern.
- Higher systolic BP, diastolic BP, and hypertension prevalence are concentrated in eastern and southeastern Austrian districts.
- Elevated rates occurred across urban, intermediate, and rural districts
- Neighboring districts tend to show similar cardiovascular risk profiles, indicating spatial clustering.
- Moran's I confirms significant positive spatial autocorrelation for hypertension prevalence ($I = 0.154$, $p = 0.025$).

LISA Cluster Findings (District Level)



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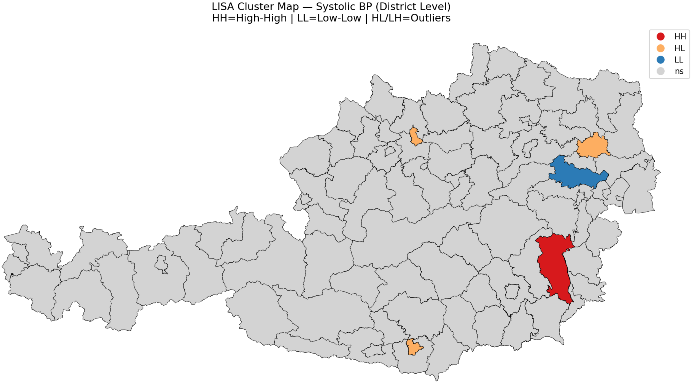
LISA Cluster Findings (District Level)



LISA Cluster Summary — Key Spatial Findings

Hypertension Rate

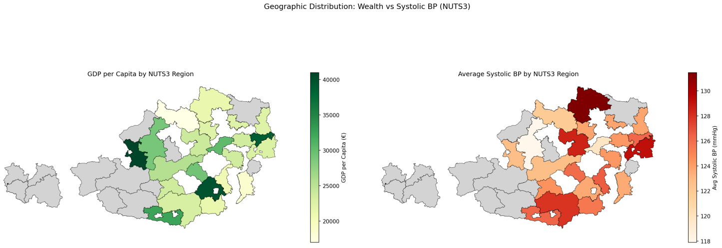
- **LL Cluster — Baden**
  - Low hypertension surrounded by low neighbouring districts
  - Stable low-risk zone
- **HL Urban Outliers**
  - Wien, Linz, Klagenfurt
  - Higher hypertension than surrounding districts

Systolic BP

- **HH Cluster**
  - Hartberg-Fürstenfeld & Neunkirchen
  - Significant southeastern high-BP cluster
- **LL Cluster — Baden**
  - Consistent low-BP spatial zone
- **HL Urban Outliers**
  - Wien, Linz, Klagenfurt

Spatial patterns suggest both regional clustering and urban outlier effects in cardiovascular risk.

Regional GDP per Capita (NUTS3) vs BP



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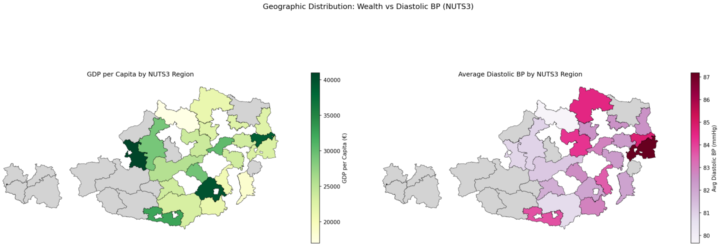
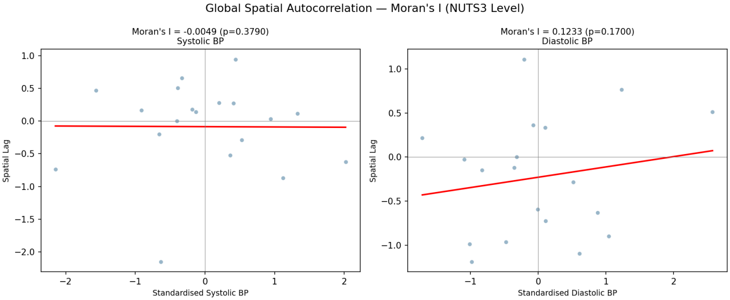


Spatial Autocorrelation of Blood Pressure (NUTS3 Level) - GDP



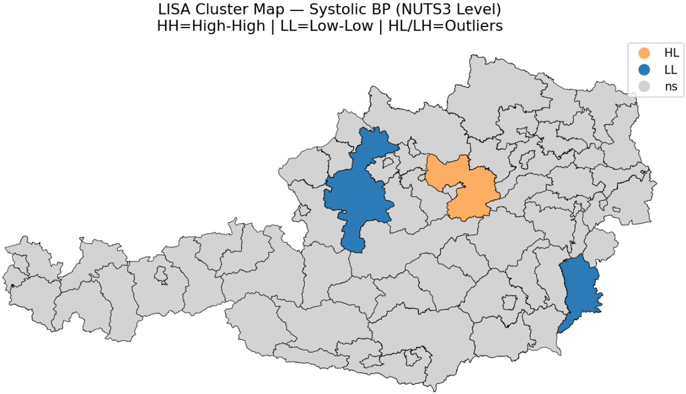
GDP and Blood Pressure — Key Findings

- Western and central Austrian regions generally showed higher GDP levels with comparatively lower or moderate blood pressure values.
- Eastern and southeastern regions exhibited higher systolic BP and hypertension prevalence despite mixed economic profiles.
- Some economically strong urban regions such as Wien and Linz-Wels also showed elevated BP levels, suggesting that higher GDP alone does not protect against cardiovascular risk.
- Correlation analysis showed only a weak relationship between GDP and systolic BP:

$$r = -0.064, \quad p = 0.784$$

- Spatial autocorrelation at the NUTS3 level was not statistically significant, indicating no strong regional clustering pattern linked to GDP.

Local Spatial Clusters of Systolic BP (NUTS3 Level)



NUTS3 Spatial Clustering — Key Findings

- Although global spatial autocorrelation is not significant at the NUTS3 level ($I_{\text{sys}} = -0.005$, $I_{\text{dia}} = 0.123$), local clustering is still observed.
- A few regions in central-western and southeastern Austria show Low-Low (LL) BP clusters.
- Some central and northeastern regions appear as High-Low (HL) outliers, indicating elevated BP relative to neighbouring regions.
- These findings suggest that BP spatial structure becomes weaker at broader regional aggregation levels compared with district-level analysis.